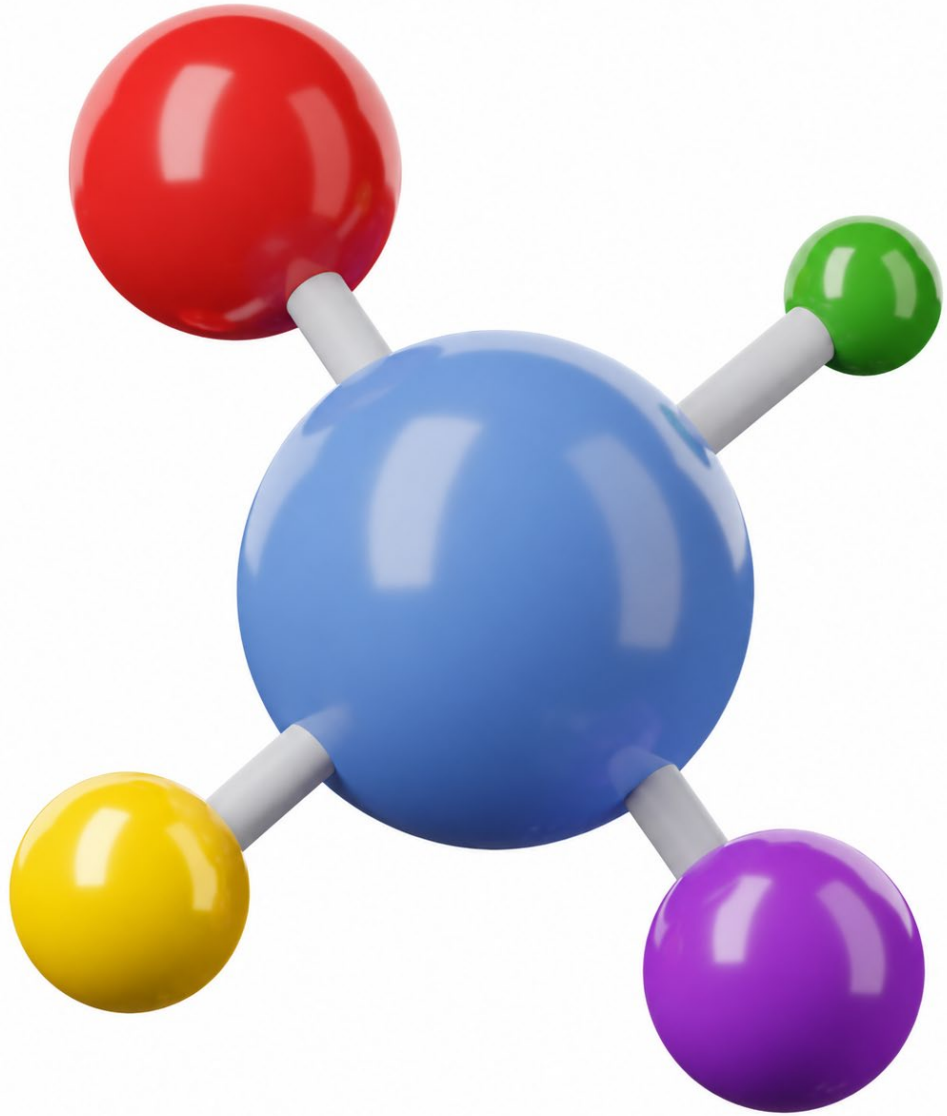


Amino acid synthesis

Nitrogen incorporation

Learning goals

- Concept of nitrogen fixation
- Incorporation of ammonia into biomolecules
- Understand the regulation of glutamine synthetase
- Overview of amino acid biosynthesis
- Essential amino acids
- Examples of molecules derived from amino acids

Biochemistry of molecular nitrogen

- N_2 constitutes 78% of Earth's atmosphere
- $N_2 + 3\ H_2 \rightarrow 2\ NH_3 \quad \Delta G^{o'} = -33.5$ kJ/mol
- Breaking the triple bond requires a high activation energy
- Few organisms can convert N_2 to biologically useful forms

Ammonia is incorporated into biomolecules through Glu and Gln

Nitrogen fixation

- $N_2 + 8\ H^+ + 8\ e^- + 16\ ATP \rightarrow NH_3 + H_2 + 16\ ADP + 16\ P_i$
- NH_3 is incorporated into glutamine by *glutamine synthetase*
- Glutamate and glutamine are the major nitrogen donors in biosynthesis

Glutamine synthetase

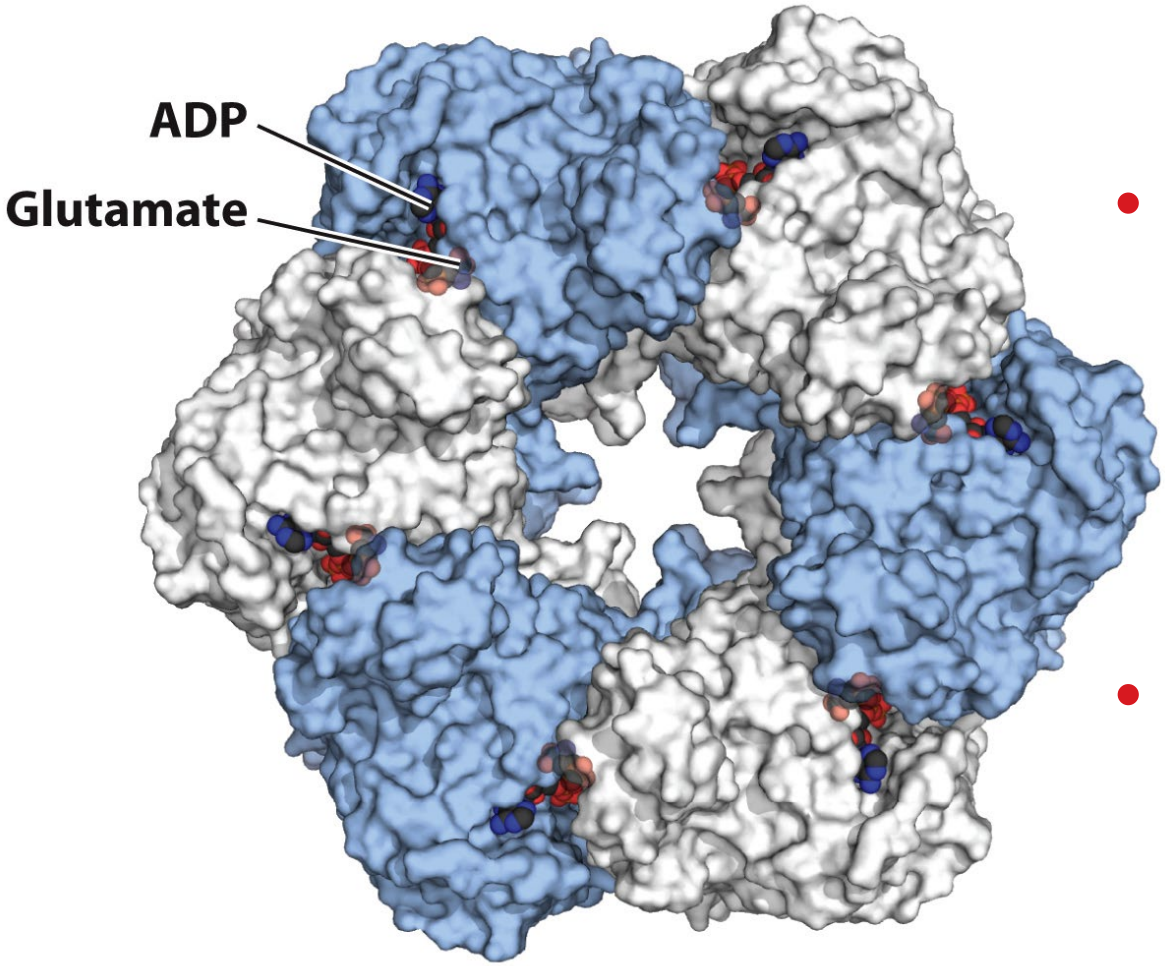


Figure 22-7
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- **Structure**
  - 12 identical subunits
  - Cumulative feedback inhibition by eight end products of glutamine metabolism
- **Regulation**
  - Allosterically and by covalent modifications

This *E. coli* enzyme provides a remarkable example of multilevel enzyme regulation

Allosteric regulation of glutamine synthetase

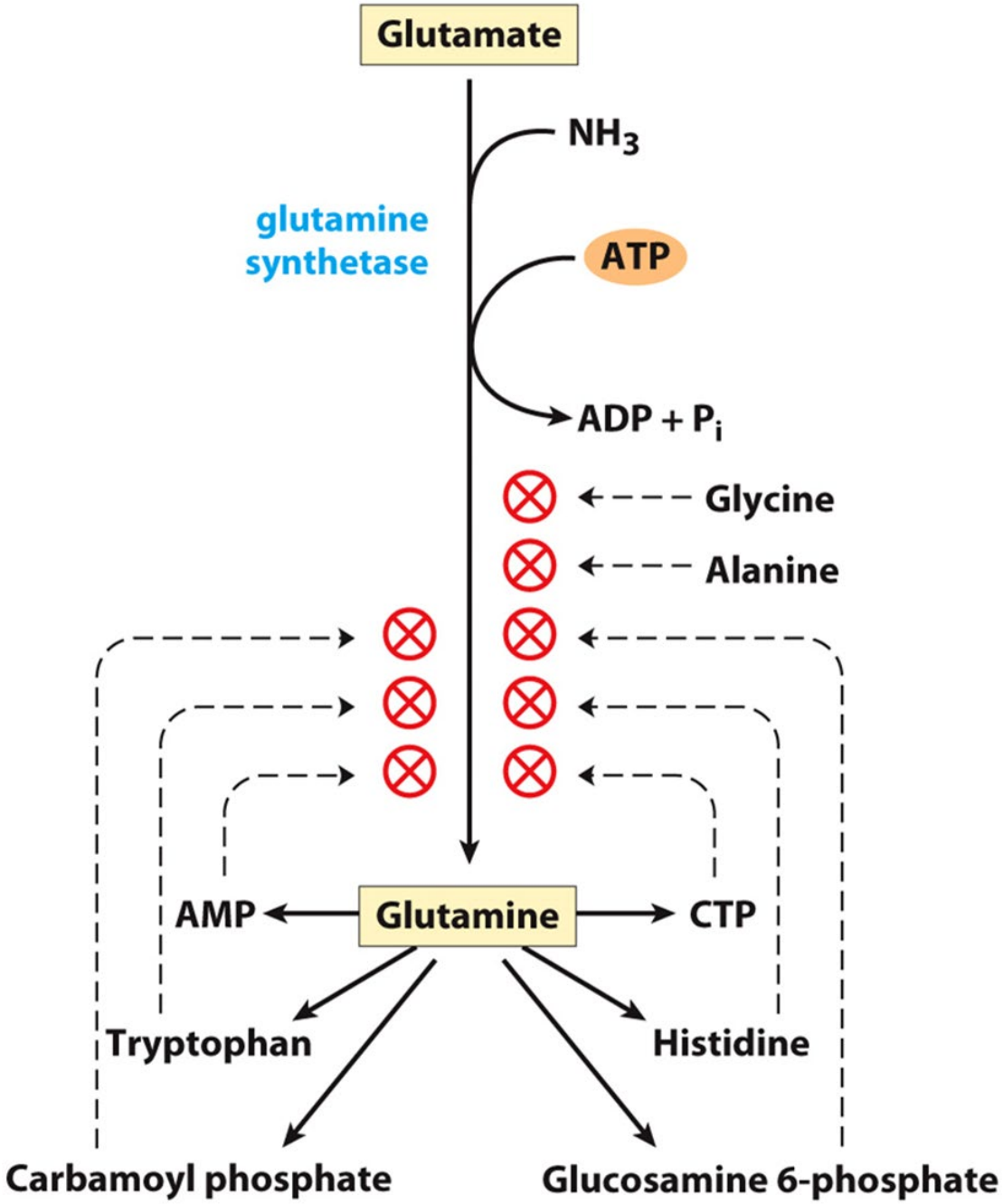


Figure 22-8
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Covalent modification of glutamine synthetase: Adenylylation

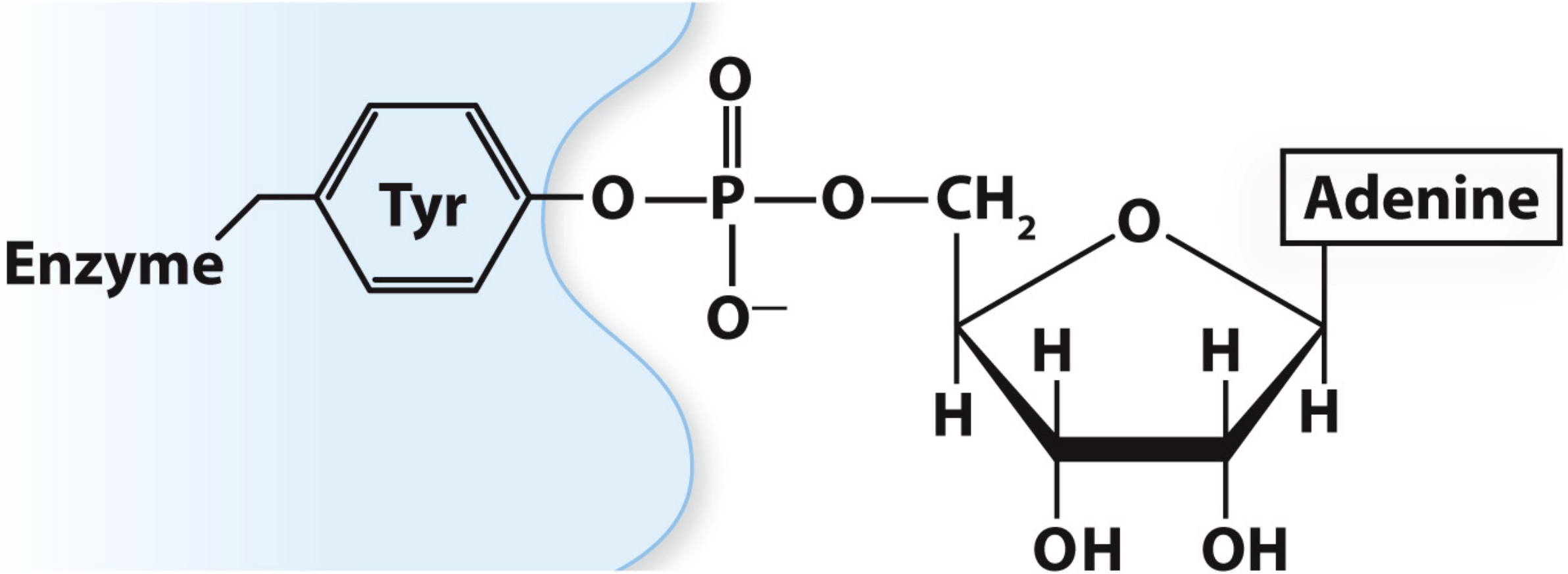


Figure 22-9a
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Regulation of glutamine synthetase

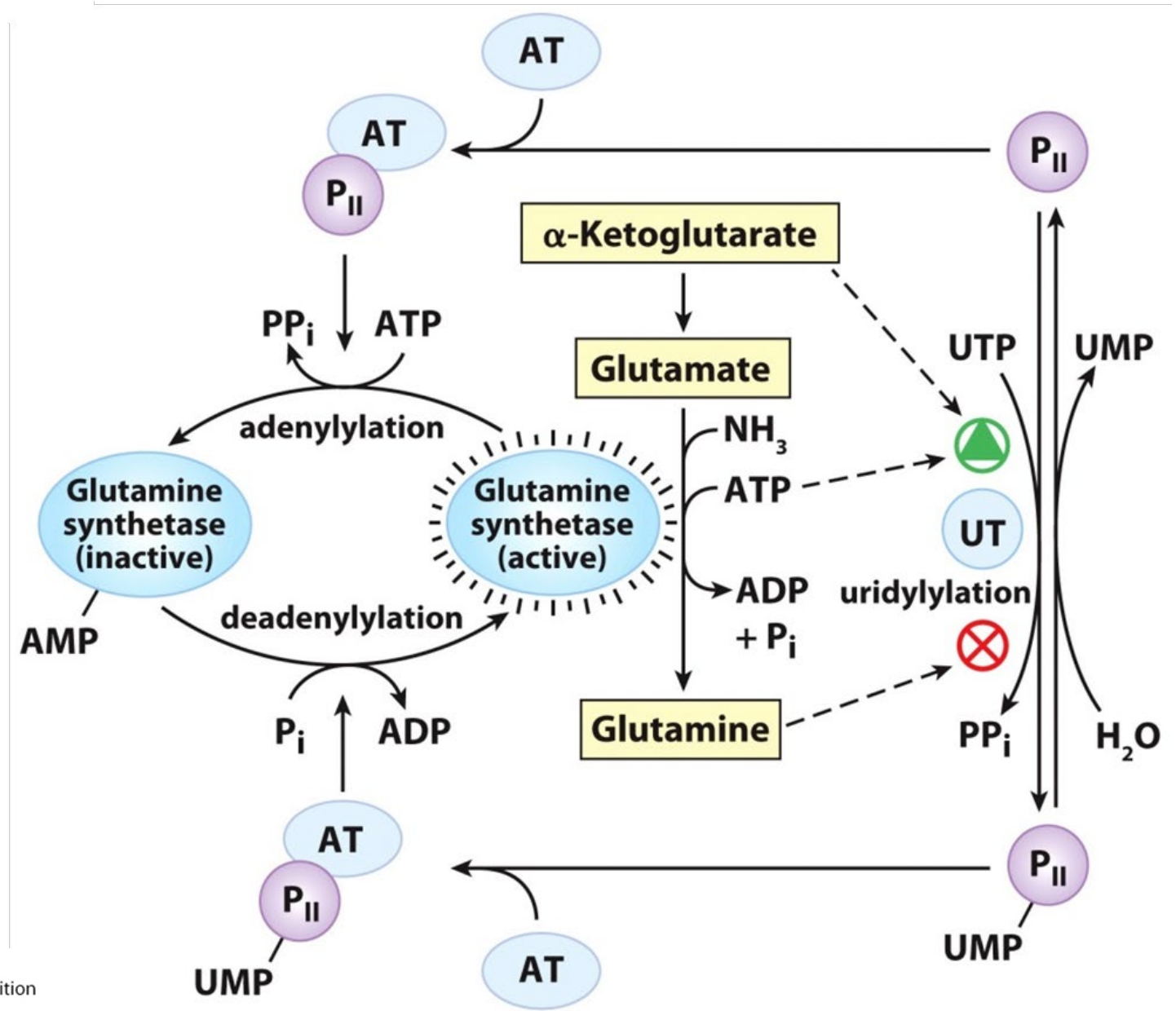


Figure 22-9b
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Amino acid synthesis

- Source of nitrogen is Glu or Gln
- Carbon skeletons derive from intermediates of:
  - Glycolysis
  - Citric acid cycle
  - Pentose phosphate pathway
- Many bacteria can synthesize all 20 standard amino acids
- Mammals must obtain essential amino acids from the diet

Amino acid synthesis: overview

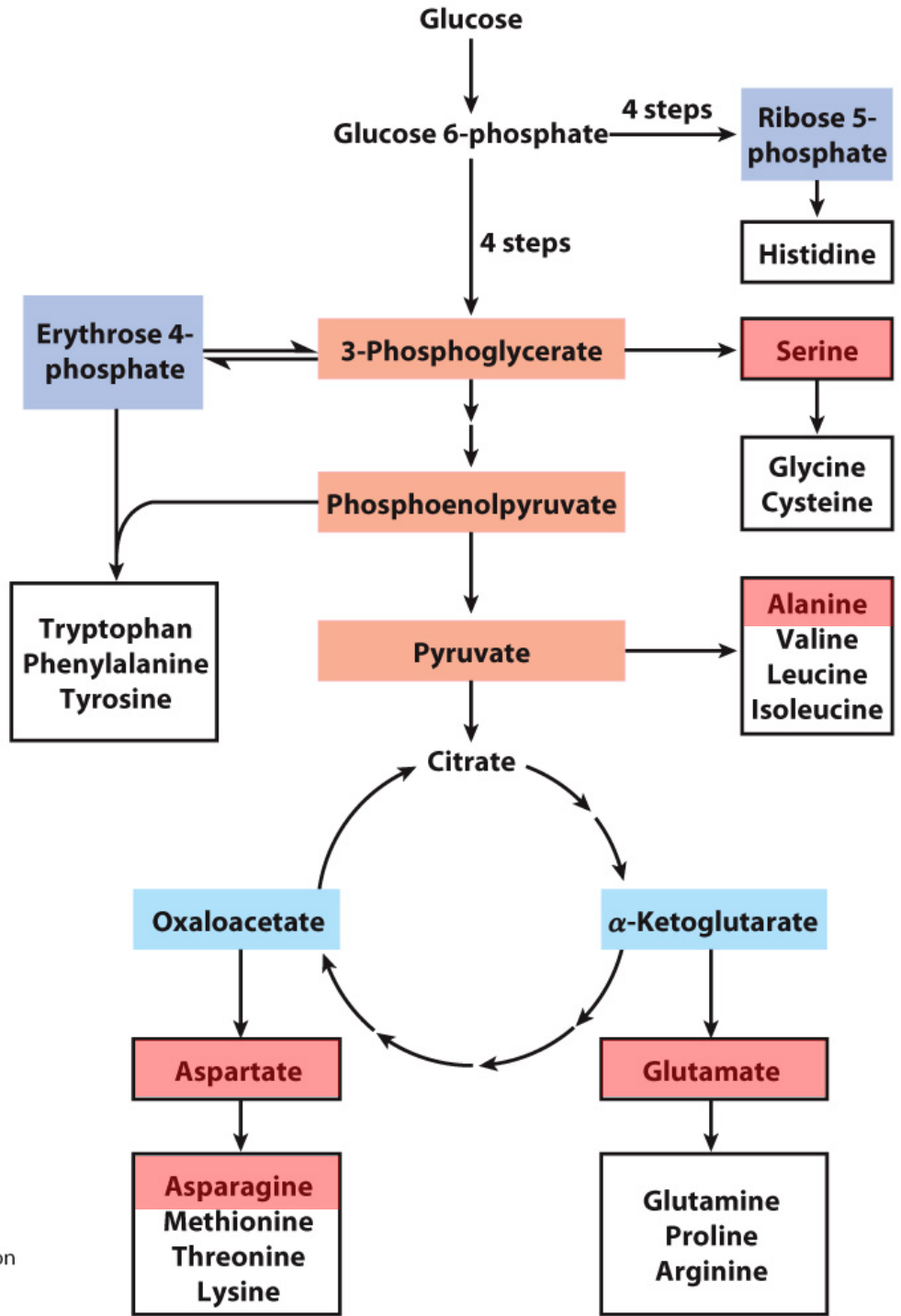


Figure 22-11
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Humans cannot synthesize all amino acids in sufficient amounts

TABLE 18-1	Nonessential and Essential Amino Acids for Humans *Lehninger 7th ed.*	
Nonessential	**Conditionally essential[a]**	**Essential**
Alanine	Arginine	Histidine
Asparagine	Cysteine	Isoleucine
Aspartate	Glutamine	Leucine
Glutamate	Glycine	Lysine
Serine	Proline	Methionine
	Tyrosine	Phenylalanine
		Threonine
		Tryptophan
		Valine

[a]Required to some degree in young, growing animals and/or sometimes during illness.

Cofactors involved in amino acid metabolism

- Pyridoxal phosphate (PLP)
- Transfer of carbon-containing groups:
  - Biotin – carboxyl groups
  - Tetrahydrofolate (THF) – one-carbon units
  - S-adenosylmethionine (adoMet) – methyl groups
- Tetrahydrobiopterin (BH_4)

Molecules derived from amino acids

- Serine → sphingosine
- Histidine → histamine
- Tyrosine → dopamine → epinephrine
- Glutamate → γ-aminobutyrate (GABA)
- Tryptophan → serotonin
- Methionine + ornithine → spermidine and spermine
- Glutamate-cysteine-glycine → glutathione
- Glycine → porphyrins → heme
- Glycine + arginine → creatine
- Arginine → nitric oxide (NO)
- Purine and pyrimidine nucleotides